

D5 (D5-EXTRA)

COMPLEX MULTIFUNCTIONAL ADMIXTURE TO CONCRETES, MORTARS, DRY BUILDING MIXES AND CEMENTS

Highly effective multifunctional admixture D5 is an innovative product developed in 2006 under a grant from the State Foundation for Innovation Support (Contract №3492p/5563, research supervisor, D.Sc. O.K. Bazoev).

Use of the admixture D5 (D5 EXTRA) allows you to:

COMPLETELY ELIMINATE ANY ADDITIONAL WATERPROOFING
for concrete and reinforced concrete structures and implement the highly
reliable waterproof technology

« W h i t e T a n k »

REDUCE CONSTRUCTION COSTS AND TIMELINES BY 15-20%

MAIN PROPERTIES OF CONCRETE WITH ADMIXTURE D5 (D5-EXTRA)

- **HIGH WATER-RESISTANCE:** Concrete with the admixture D5 (D5 EXTRA) achieves the highest possible water resistance (up to 20 bar and more), eliminating the need for any additional waterproofing in water-saturated soils, water environment or directly in water. **This high water-resistance of concrete with D5 (D5 EXTRA) is maintained throughout the structure's entire service life.** Furthermore, using D5 (D5-EXTRA) in plaster eliminates dampness and mold in high-humidity areas.
- **HIGH SULFATE-RESISTANCE:** Concrete with the admixture D5 (D5 EXTRA) provides high sulfate-resistance even with ordinary Portland cement. This eliminates the need for additional waterproofing of concrete even when concrete is exposed to aggressive environments, including sea water, oil and petroleum products.
- **HIGH STRENGTH:** The use of D5 (D5 EXTRA) increases concrete strength at the age of 28 days by an average of 30% compared to concrete without admixtures (using concrete mixes with equal slump), or alternatively cement content can be reduced by 10-15%.
- **DOUBLING THE RATE OF STRENGTH GAIN** reduces construction time by at least 15%, while enhancing the quality and durability of the structures.
- **HIGH FROST-RESISTANCE:** The use of D5 (D5 EXTRA) increases the frost-resistance of concrete by at least 2 classes and enables the production of concrete with frost-resistance of F400 and more.
- **PLASTICIZING CHARACTERISTICS:** The admixture D5 (D5 EXTRA) increases the flowability of concrete mixes from P1 to P5 or reduces water consumption by 15-20% in mixes with equal slump. The use of D5 (D5 EXTRA) eliminates the need for any other plasticizers.
- **SELF-SEALING OF THROUGH CRACKS** up to 0.5 mm wide, which may develop in structures during operation due to dynamic loads, including seismic activity. A necessary condition is water filtration through the cracks. Under normal conditions, this property ensures that «cold» joints in concrete using D5 (D5 EXTRA) do not leak under the water pressure of up to 12 bar.
- **ECO-FRIENDLINESS:** The admixture D5 (D5 EXTRA) contains no components that negatively affect cement stone or steel reinforcement. It does not form toxic compounds in the atmosphere or water. **The admixture D5 (D5 EXTRA) is approved for use in concrete and reinforced concrete structures that are in contact with drinking water.**

HS CODE: 3824 40 000 0

D5 (D5-EXTRA)

RECOMMENDATIONS FOR USE

1. Before using the admixture D5 (D5 EXTRA) at precast concrete plants or mixing stations, it is necessary to conduct preliminary laboratory tests to determine its effectiveness with the specific cements and aggregates used at that particular plant.

*It is recommended to test two concrete mixes with equal flowability (slump): a **Control Mix** (without additives) and a **Basic Mix** containing 3% of D5 (D5 EXTRA) by weight of cement. Both mixes must be produced using the same components, with the same dosage and with the same slump. To achieve equal slump of both mixes, the water content in the Basic mix should be reduced by an average of 15-25% compared to the Control Mix.*

2. When producing the concrete mix, the admixture D5 (D5 EXTRA) is added in the concrete mixer in dry form along with the sand or cement. **Pre-mixing the admixture D5 (D5 EXTRA) with water is not required!** The mixing time is 60-90 seconds, which ensures even distribution of D5 (D5 EXTRA) throughout the mixture. It is important to note that the admixture D5 (D5 EXTRA) contains a high-performance dry superplasticizer, which must be accounted for when determining the water content of the mix.

Transportation, pumping and placement in formwork of the concrete mix with the admixture D5 (D5 EXTRA) require no special procedures. However, the concrete mix with D5 (D5 EXTRA) does not segregate, retains its flowability longer, exhibits improved pumpability and flows easily when vibrated.

3. To provide retention of flowability of concrete mixes for 2 hours or more, and to produce self-compacting concrete, it is necessary to produce a concrete mix with the admixture D5 EXTRA with an initial slump of 230-250 mm, while the dosage of the admixture should be 4% by the weight of cement.

**The recommended dosage of the admixture D5 (D5-EXTRA) is
2-3% by the total weight of cement**

4. Almost all monolithic construction projects face the issue of «cold» joints, formed at locations where concreting is interrupted, i.e., when previously laid concrete has already hardened and a fresh concrete mix is poured onto it. These joints are the primary points for future water infiltration.

In concrete containing D5 (D5 EXTRA) cold joints will not leak. To ensure this, the following steps must be taken:

- the surface of the previously poured concrete with the admixture D5 or D5-EXTRA should be cleaned of cement crust immediately after hardening (6-8 hours after pouring) at the location of the future «cold» joint. It is necessary to expose the aggregate, wash away dust and debris, and thoroughly moisten the concrete at the joint area;

- after breaks in concreting (lasting days to weeks), just before pouring freshly-mixed concrete with D5 (D5 EXTRA), the cold joint surface must be re-cleaned to remove dust and debris from the surface of earlier stripped concrete;

- under these conditions, D5 (D5 EXTRA) ensures high adhesion and molecular-level fusion between hardened and fresh concrete, providing superior waterproofing for cold construction joints (both vertical and horizontal).